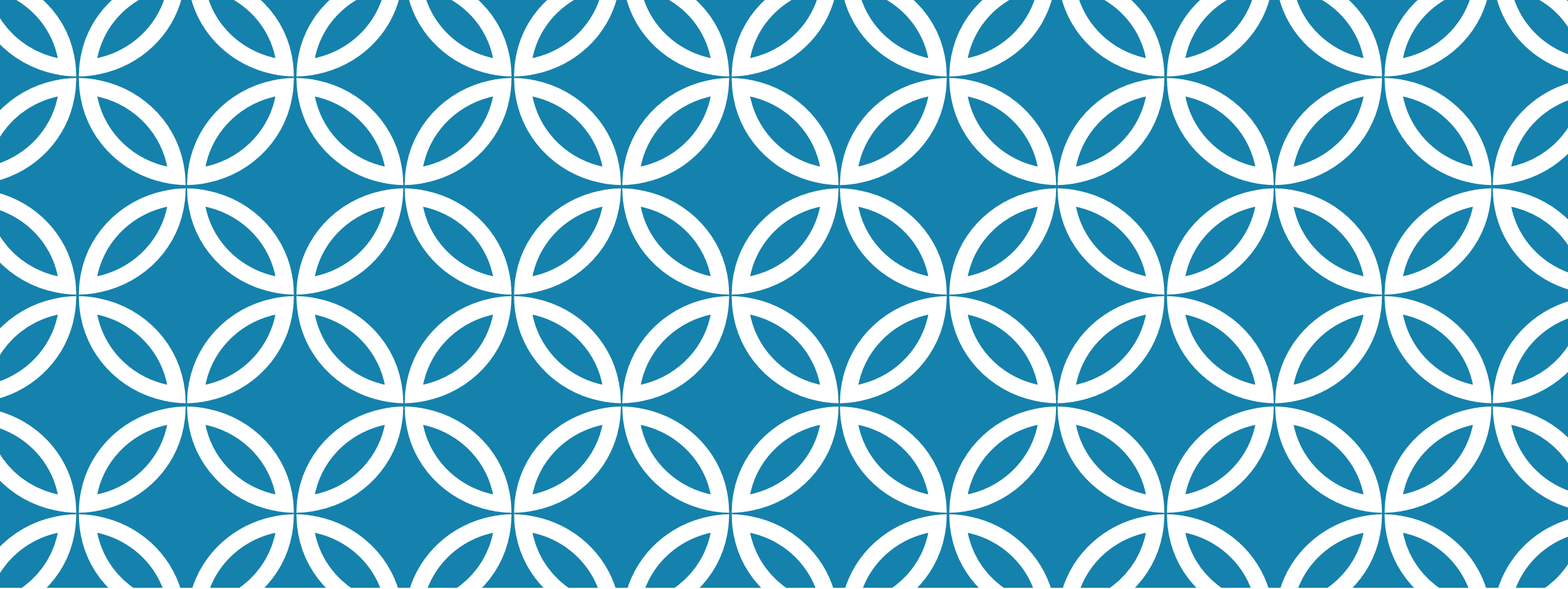




Relations and Functions

Logic in Computer Science Lab
No. 2
Gyöngyi Bujdosó
2025

René Descartes

He was a French philosopher, scientist, and mathematician.

His name in Latin: Renatus Cartesius

He was widely considered a seminal figure in the emergence of modern philosophy and science. Mathematics was paramount to his method of inquiry, and he connected the previously separate fields of geometry and algebra into analytic geometry.

Pronunciation: /deɪˈkɑːrt/ day-KART or UK: /ˈdeɪkɑːrt/ DAY-kart; French: [ʁəne dekaʁt]



René Descartes
(1596–1650)

Relations

$A \times B$: is the Cartesian product of A and B

(read: A cross B , or A times B) where A and B are nonempty sets

$= \{\langle x, y \rangle \mid x \in A, y \in B\}$, so $\langle x, y \rangle$ is an ordered pair.

n -ary Cartesian product:

If we have $n (n \geq 3)$ nonempty sets $S_1, S_2, \dots, S_n$, then the Cartesian product of the sets $S_1, S_2, \dots, S_n$ is the following set:

$$S_1 \times S_2 \times \dots \times S_n = \{\langle x_1, x_2, \dots, x_n \rangle \mid x_1 \in S_1, x_2 \in S_2, \dots, x_n \in S_n\},$$

where $\langle x_1, x_2, \dots, x_n \rangle$ is an ordered n -tuple.

Relations

Any subset of an n -ary Cartesian product is called a **relation**.

If $R \subset A \times A \times \cdots \times A$ then R is called as an **n -ary relation over A** .

If $n = 2$, this is a **binary relation**.

Denotations: $(x, y) \in R$ or $R(x, y)$

Properties of Binary Relations

With a binary relation, we will define relations between elements of two sets. When $(h_1, h_2) \in R$ (h_1 is related to h_2) then we use the notation: $h_1 R h_2$. The $R \subseteq H \times H$ is a binary relation on H .

A binary relation is

- a reflexive relation, if $h R h$ for all $h \in H$,
- an anti-reflexive relation, if there is no $h \in H$ where $h R h$,
- a symmetric relation, if (for all $h_1, h_2 \in H$) when $h_1 R h_2$ then $h_2 R h_1$,
- an anti-symmetric relation: if there is no $h_1 R h_2$ such that $h_2 R h_1$, unless $h_1 = h_2$,
- a transitive relation, if (for all $h_1, h_2, h_3 \in H$) when $h_1 R h_2$ and $h_2 R h_3$ then $h_1 R h_3$

Properties of Binary relations

Not every relation which is not reflexive is anti-reflexive.

We can define relations where some elements are related to themselves, but others are not (that is, neither all nor none are).

Example

The binary relation

“the product of x and y is even”

- **reflexive** on the set of **even** numbers,
- **anti-reflexive** on the set of **odd** numbers, and
- **neither reflexive nor anti-reflexive** on the set of **natural** numbers

Relation properties & Functions

Let R be a binary relation on H ($R \subseteq H \times H$)

- R is called **injective** (left unique) relation if, for all $x \in H$ and $y \in H$:

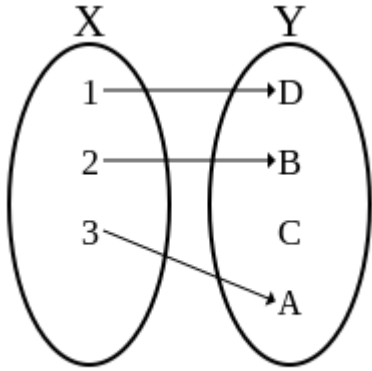
$$xRz \text{ and } yRz \implies x = y$$

- R is called **surjective** if, for all $y \in H$ there exists an $x \in H$ where xRy .
It's also called **right total** because $\text{Rng}(R) = H$.
- R is called **left total** if $\text{Dom}(R) = H$.
- R is called **bijective** when, it is both injective and surjective.
- R is called **functional** (right unique) or **partial function** if, for all $x, y, z \in H$:

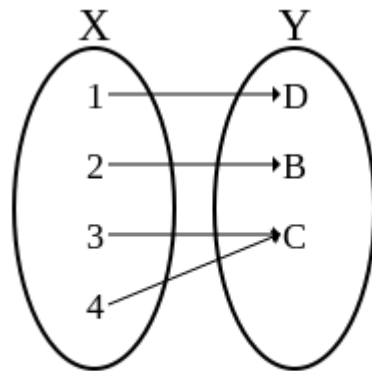
$$xRy \text{ and } xRz \implies y = z$$

- R is called **function**, if it is **functional** and **left total**.

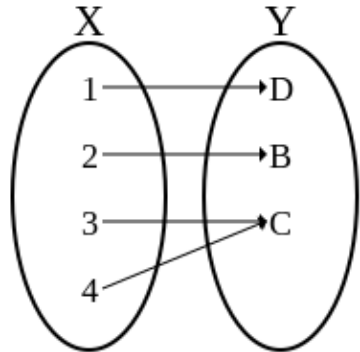
Injective, surjective, and bijective relations



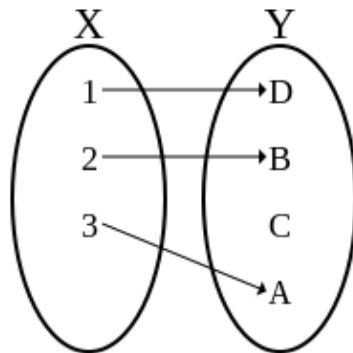
Injective relation



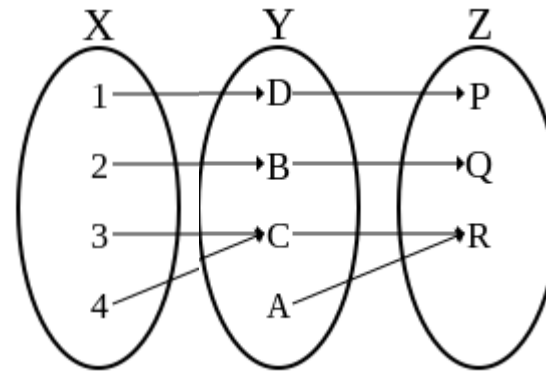
Not injective relation



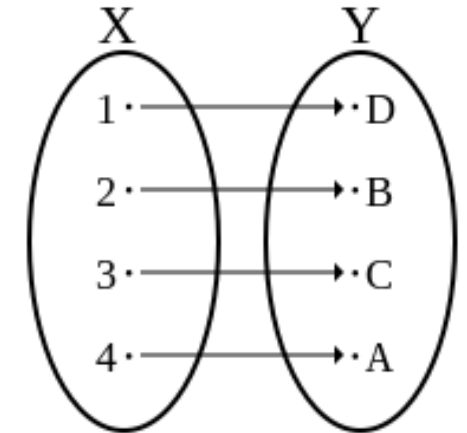
Surjective relation



Not surjective relation

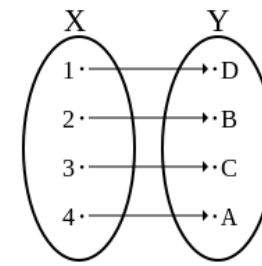


Sequence of surjective relations:
the last relation has to be
surjective

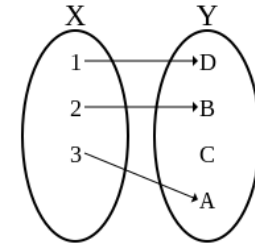


Injective and surjective, i.e.
bijective relation

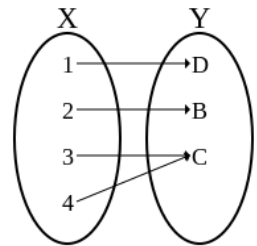
Examples



Injective and surjective



Injective
relation



Not injective
relation

Injective Relations

1. We use the set of positive whole numbers ($\mathbb{Z}_+$), and $x \in \mathbb{Z}_+$.

The mapping $f: x \rightarrow 2x$, i.e. the relation:

$R = \{(x, 2x) \mid x \in \mathbb{Z}_+\}$ has the elements

$R = \{(1,2), (2,4), (3,6), (4,8), (5,10), \dots\}$

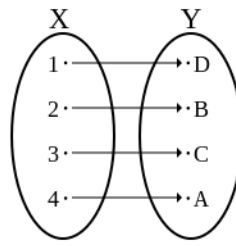
is injective.

2. $x \in \mathbb{N}$, $f: \mathbb{N} \times \mathbb{N}$, $f: x \rightarrow x + 1$

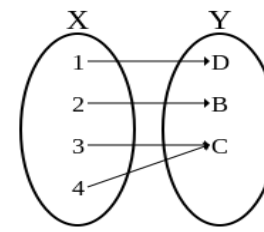
Not Injective

1. $f: \mathbb{N} \times \mathbb{N}$, $x \in \mathbb{N}$, $f: x \rightarrow |x - 1| + 1$
2. $f: \mathbb{R} \times \mathbb{R}$, $x \in \mathbb{R}$, $f: x \rightarrow x^n - x$

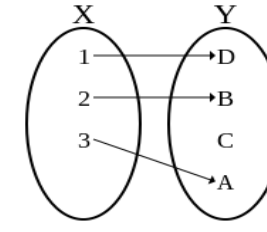
Examples



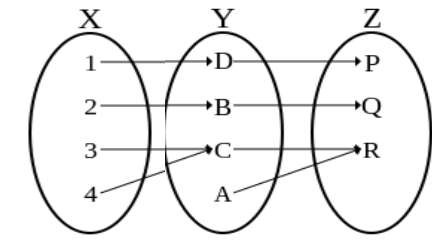
Injective and surjective



Surjective relation



Not surjective relation



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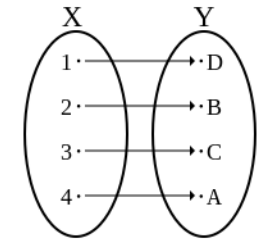
Surjective

1. $x \in \mathbb{R}, f: \mathbb{R} \times \mathbb{R}, f: x \rightarrow 2x + 1$
2. Given that the
set $A = \{1, 2, 3\}$,
set $B = \{4, 5\}$ and let
the relation $R: A \times B$,
 $R = \{(1, 4), (2, 5), (3, 5)\}$
that is surjective.
3. $x \in \mathbb{R}, f: \mathbb{R} \times [0, \infty), f: x \rightarrow |x|$

Not Surjective

1. $x \in \mathbb{N}, f: \mathbb{N} \times \mathbb{N}, f: x \rightarrow x + 1$
2. $x \in \mathbb{R}, f: \mathbb{R} \times \mathbb{R}, f: x \rightarrow |x|$

Examples



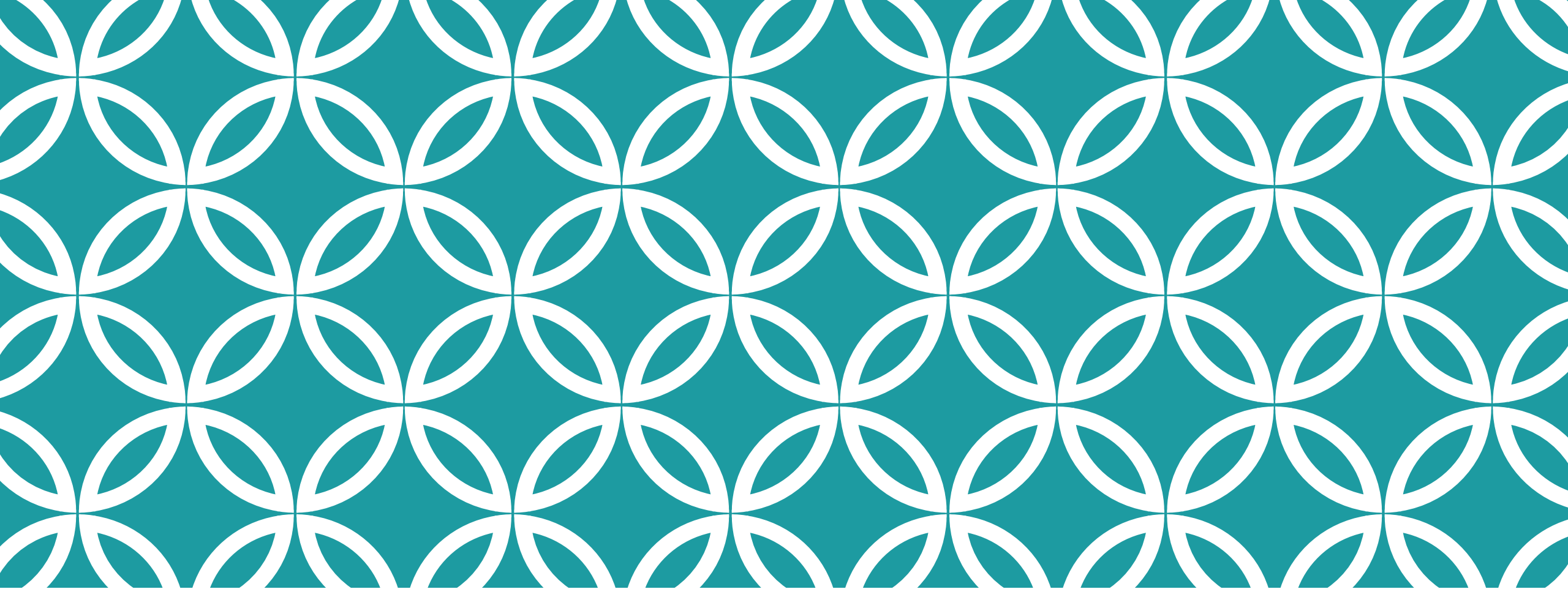
Injective and surjective, i.e.
bijective

Bijjective

1. $x \in \mathbb{R}, f: \mathbb{R} \times \mathbb{R}, f: x \rightarrow 2x + 1$
2. $x \in \mathbb{R}, f: \mathbb{R} \times \mathbb{R}, f: x \rightarrow x$
3. $x \in [0, \infty), f: [0, \infty) \times [0, \infty), f: x \rightarrow x^2$

Not Bijjective

1. $x \in \mathbb{N}, f: \mathbb{N} \times \mathbb{N}, f: x \rightarrow x + 1$
2. $x \in \mathbb{R}, f: \mathbb{R} \times \mathbb{R}, f: x \rightarrow e^x$
3. $x \in \mathbb{R}, f: \mathbb{R} \times \mathbb{R}, f: x \rightarrow x^2$



Exercises

Binary Relations

With a binary relation, we will define relations between elements of two sets. When $(h_1, h_2) \in R$ (h_1 is related to h_2) then we use the notation: $h_1 R h_2$. The $R \subseteq H \times H$ is a binary relation on H .

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Exercises

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1. Which properties hold for the relation $\in$ over finite sets?
2. Which properties hold for the relation $\subseteq$ over sets?
3. Which properties holds for the relation *perpendicular* over lines?
4. Which properties holds for the relation *parallel* over lines?
5. Which properties holds for the relation *divisible* over positive whole numbers? x divisible by y ($y|x$), if there exists z such that $y \cdot z = x$.

Exercises

A) a reflexive relation, if hRh for all $h \in H$,

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E) a transitive relation, if (for all $h_1, h_2, h_3 \in H$) when h_1Rh_2 and h_2Rh_3 then h_1Rh_3

1. Which properties hold for the relation $<$ over real numbers?
2. Which properties hold for the relation $\leq$ over real numbers?
3. Which properties hold for the relation $\neq$ over real numbers?

Functions

Let R be a binary relation on H ($R \subseteq H \times H$)

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- R is called **function**, if it is **functional** and **left total**.

1. Which of the following relations are functions?

(a) $A = \{1, 2, 4\}, B = \{3, 6, 12\}, f_1 \subseteq A \times B, x f_1 y \iff xy = 12$

(b) $f_2 \subseteq \mathbb{N} \times \mathbb{N}, x f_2 y \iff x = y$

(c) $f_3 \subseteq \mathbb{N} \times \mathbb{N}, x f_3 y \iff x < y$

(d) $f_4 \subseteq \mathbb{N} \times \mathbb{N}, x f_4 y \iff 2x = 3y$

(e) $f_5 \subseteq \mathbb{N} \times \mathbb{N}, x f_5 y \iff x|y$